Properties of real numbers

1 Determine whether the following number sentences are true or false for every value of x.

a $x+1=1+x$

b $x\times 3=3\times x$

c $-5+x=x-5$

d $x-2=-2+x$

e $1\times 8x=18x$

f $(3+7)+x=3+(7+x)$

g $(-2+x)+3=-2+(x+3)$

h $(x\times 7)\times 3=x\times(7\times 3)$

i $9x\times\frac{1}{9}=x$

j $\frac{1}{6}x\times 6=x$

k $2x=x-x+x$

2 Is this number sentence true or false for every positive value of v?

$$v\div 3=3\div v$$

3 Is this number sentence true or false if a is not 9?

$$a-9=9-a$$

4 Determine whether the following number sentences are true or false for every value of a and b.

a $3ab\times 1=3ab$

b $0\times 4ab=0$

c $a+b=b+a$

d $ab=ba$

5 For each of the following, find the missing term that would make the number sentence true.

a $0+\square=9m$

b $1\times\square=4b$

c $\square-0=3d$

d $a\times 19=19\times\square$

e $d+\frac{1}{7}=\square+d$

6 Using the commutative property of addition to complete the following:

a $5\,(s+6)=\square\,(\square+\square)$

b $8\,(4+m)=\square(\square+\square)$

7 Consider the two expressions $x - 5$ and $5 -$
Vanessa substituted $x = 5$ into each expression to determine whether they are equivalent.
Determine whether the following statements are true.

a The expressions will have different values, so therefore they are not equivalent.

b Both expressions will have the same value, so therefore they are equivalent.

c Both expressions will have the same value, but Vanessa needs to test again with another value of x.

d The expressions will have different values, but Vanessa needs to test again with another value of x.

8 State the reciprocal of $2x$.

9 Identify which property of addition is demonstrated by each of the following statements.

a $5 + 14 = 14 + 5$

b $32 + 5 = 5 + 32$

c $12 + (-76) = -76 + 12$

d $(5 + 7) + 2 = 5 + (7 + 2)$

e $4 + (51 + 2) = (4 + 51) + 2$

f $-7 + (9 + 8) = (-7 + 9) + 8$

g $0 + 34 = 34$

h $-3 + 3 = 0$

i $4 + (-4) = 0$

j $176 + (-176) = 0$

k $5.66 + 0 = 5.66$

l $-2.35 + 0 = -2.35$

m $(4 + 6) + 5 = 5 + (4 + 6)$

n $(-2 + 4) + 6 = 6 + (-2 + 4)$

o $x + 8 = 8 + x$

p $(2 + x) + 3 = 2 + (x + 3)$

10 Identify which property of multiplication is demonstrated by each of the following statements.

a $5\,(8 + 9) = (8 + 9)\,5$

b $4 \times (9 \times 5) = (4 \times 9) \times 5$

c $\dfrac{3}{8} \times \dfrac{8}{3} = 1$

d $5 \times (2 \times 7) = (5 \times 2) \times 7$

e $-9 \times (2 \times 3) = (-9 \times 2) \times 3$

f $3 \times (7 \times 6) = (3 \times 7) \times 6$

g $8 \times 7 = 7 \times 8$

h $(3 \times y) \times 11 = 3 \times (y \times 11)$

i $-\dfrac{3}{4} = -\dfrac{3}{4} \times \dfrac{5}{5} = -\dfrac{15}{20}$

11 By which property are $9x = -27$ and $\dfrac{9x}{9} = \dfrac{-27}{9}$ equivalent equations?

12 Determine whether the following statements demonstrate the distributive property.

a $8(u+v)=8u+8v$

b $2(5x)+2(3y)=2(5x+3y)$

c $3(4u)-3(5v)=3(4u-5v)$

13 For each of the following:

i Identify which property needs to be used to complete the statement.

ii Complete the statement by finding the missing value.

a $-7+8=8+\square$

b $-6\times 5=\square\times(-6)$

c $(9+7)+6=9+(\square+6)$

d $6\times(9\times 7)=(6\times\square)\times 7$

14 Rewrite the following expressions using the commutative property.

a $x+3$

b $4\times q$

c $g\times d$

15 Complete the following statements using the associative property of multiplication.

a $2\times(a\times b)=(\square\times\square)\times\square$

b $(ab)c=\square(\square)$

c $9+(a+b)=(\square+\square)+\square$

16 Complete the following statement using the associative property of multiplication.

$$\begin{aligned}8(5h)&=(\square\times 5)h\\&=\square h\end{aligned}$$

17 Rewrite $3(6+5)$ using the distributive property. Do not evaluate the expression.

18 Consider the statement:

$$a+(-a)=0$$

Explain why this is true for all values of a.

19 Consider the following statement:
'Using the commutative property of addition, we can rewrite $6x+2$ as $2x+6$.'
Is this statement true or false? Explain your answer.

20 When asked to simplify the expression $7 + 6(8x + 3)$, a student provides the following work and gets the correct answer.

$$
\begin{aligned}
&(1)\quad 7+6\,(8x+3) = 7+(48x+18)\\
&(2)\quad = 7+(18+48x)\\
&(3)\quad = (7+18)+48x\\
&(4)\quad = 25+48x\\
&(5)\quad = 48x+25
\end{aligned}
$$

Identify the algebraic property used in:

a Step 1 b Step 2 c Step 3 d Step 5

Additional questions

21 State whether each expression is an example of the associative law, commutative law, reordering or distributive law.

a $15 + 5 = 5 + 15$

b $24 \times 6 = 6 \times 24$

c $54 \div 9 = 9 \div 54$

d $24 + 6 + 2 = 24 + (6 + 2)$

e $126 \div 3 = (120 + 6) \div 3$

f $(15 + 6) \times 4 = (15 \times 4) + (6 \times 4)$

22 State whether the following statements are true or false.

a $4 \times (3 + 5) = 4 \times 3 + 4 \times 5$

b $4 + (3 \times 5) = 4 \times 3 + 4 \times 5$

c $(4 \times 3) + 5 = 4 \times 5 + 3 \times 5$

d $(4 + 3) \times 5 = 4 \times 5 + 3 \times 5$

23 Fill in the boxes to complete the working out using the associative law.

a $35 + 29 + 11 = 35 + (\square + \square)$

b $28 + 46 - 16 = 28 + (\square - \square)$

24 Consider $27 + 48 + 13$.

a Which pair of numbers will be easiest to add together first?

b Fill in the boxes to complete the working out.

$27 + 48 + 13 = 27 + (\square + \square)$

25 Fill in the boxes to complete the working out using the associative law.
$13 \times 4 \times 5 = \square \times (4 \times \square)$

26 Evaluate $17 + 39 + 23$ using reordering.

27 Consider $52 - 24 - 12$.

a Is $52 - 12 - 24$ equal to the above expression?

b Which of the arithmetic rules can we apply to the expression to transform it into $52 - 12 - 24$?

28 Derek is evaluating the subtraction below. He noticed that it would be easier to subtract 27 from 47 first and rearranged the numbers using reordering.
Complete the following statement and then evaluate it:
$47 - 18 - 27 = \square - 27 - \square$

29 Consider $56 - 18 - 26$.

a Which of the two numbers will be easier to subtract from 56 first?

b Complete the following statement and then evaluate it:
$56 - 18 - 26 = 56 - \square - \square$

30 Evaluate $57 - 29 - 17$ using reordering.

31 Consider $48 \div 8 \div 2$.

a Is $48 \div 2 \div 8$ equal to the above expression?

b Which of the arithmetic rules can we apply to the expression to transform it into $48 \div 2 \div 8$?

32 Consider $4 \times 13 \times 5$.

a Which pair of numbers will be easiest to multiply together first?

b Complete the following statement and then evaluate it:
$4 \times 13 \times 5 = \square \times \square \times 13$

33 Evaluate $4 \times 17 \times 5$ using regrouping.

34 Consider 35×23.

a. Complete the following statement:
35×23 is the same as $35 \times (20 + \square)$.

b. Complete the following statement:
$35 \times (20 + 3)$ is the same as $35 \times \square + 35 \times 3$.

c. Which arithmetic rule explains the equality between 35×23 and $35 \times 20 + 35 \times 3$?

d. Evaluate the expression.

35 Consider 26×29.

a. Complete the following statement:
26×29 is the same as $26 \times (\square - 1)$.

b. Complete the following statement:
$26 \times (30 - 1)$ is the same as $26 \times 30 - 26 \times \square$.

c. Which arithmetic rule explains the equality between 26×29 and $26 \times 30 - 26 \times 1$?

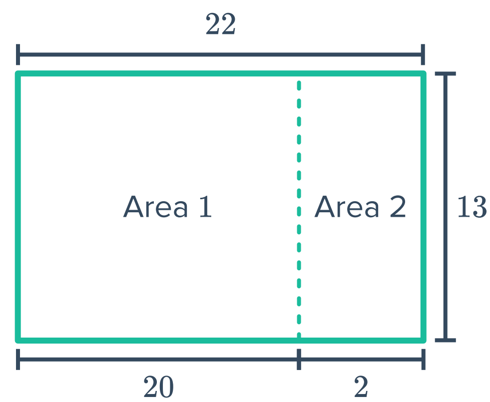
d. Evaluate the expression.

36 Consider the following rectangle:

a. Find the area of Area 1.

b. Find the area of Area 2.

c. Find the total area of the rectangle.

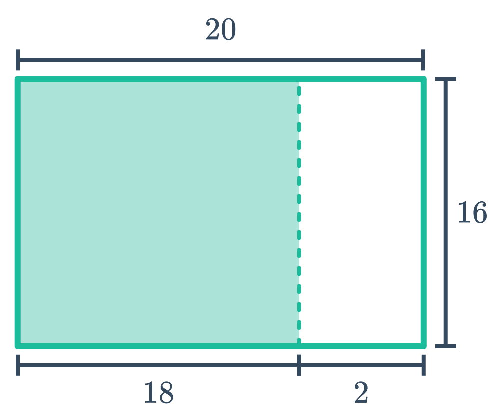


37 Consider the rectangle below:

a. Find the total area of the rectangle.

b. Find the non-shaded area of the rectangle.

c. Find the shaded area of the rectangle.



38 Evaluate the following using the distributive

a 9×998

b 7×1002

c 11×124

d 6×412

39 Dave is evaluating $132 \div 11 \div 4$. He noticed that it would be easier to divide 132 by 4 first and rearranged the numbers using reordering.
Fill in the boxes to complete his working out.
$132 \div 11 \div 4 = 132 \div \square \div 11$

40 Consider $168 \div 7$.

a Fill in the box to complete the sentence.
$168 \div 7 = (140 + \square) \div 7$.

b Fill in the box to complete the sentence.
$(140 + 28) \div 7$ is the same as $140 \div 7 + \square \div 7$.

c Which arithmetic rule explains the equality between $168 \div 7$ and $140 \div 7 + 28 \div 7$?

d Evaluate the expression.

41 Consider $119 \div 7$.

a Fill in the box to complete the sentence.
$119 \div 7$ is the same as $(140 - \square) \div 7$.

b Fill in the box to complete the sentence.
$(140 - 21) \div 7$ is the same as $140 \div \square - 21 \div \square$.

c Which arithmetic rule explains the equality between $119 \div 7$ and $140 \div 7 - 21 \div 7$?

d Evaluate the expression.

42 Evaluate the following using the distributive law.

a $1616 \div 8$

b $396 \div 4$

c $522 \div 6$

d $585 \div 13$